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(54) **MODULAR LOOM FOR TRIANGULAR OR
RECTANGULAR WEAVING**

(52) **U.S. Cl.**

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(57)

ABSTRACT

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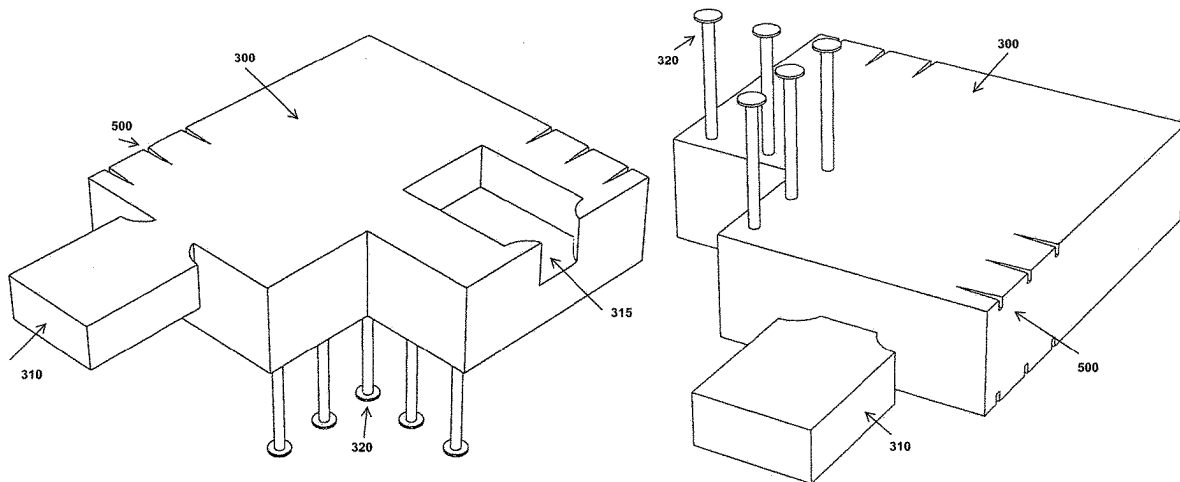
The disclosure presents a modular pin loom system for continuous strand and/or traditional weaving utilizing interlocking molded plastic modules with embedded pins to construct customizable triangular or rectangular looms. The loom dimensions are readily adjustable from 12 inches up to 8 feet on each side using combinations of generic side modules matched with 45 or 90 degree angle end pieces to transform shaping. The flanged modules securely interconnect flush with one another to maintain loom integrity without distortion.

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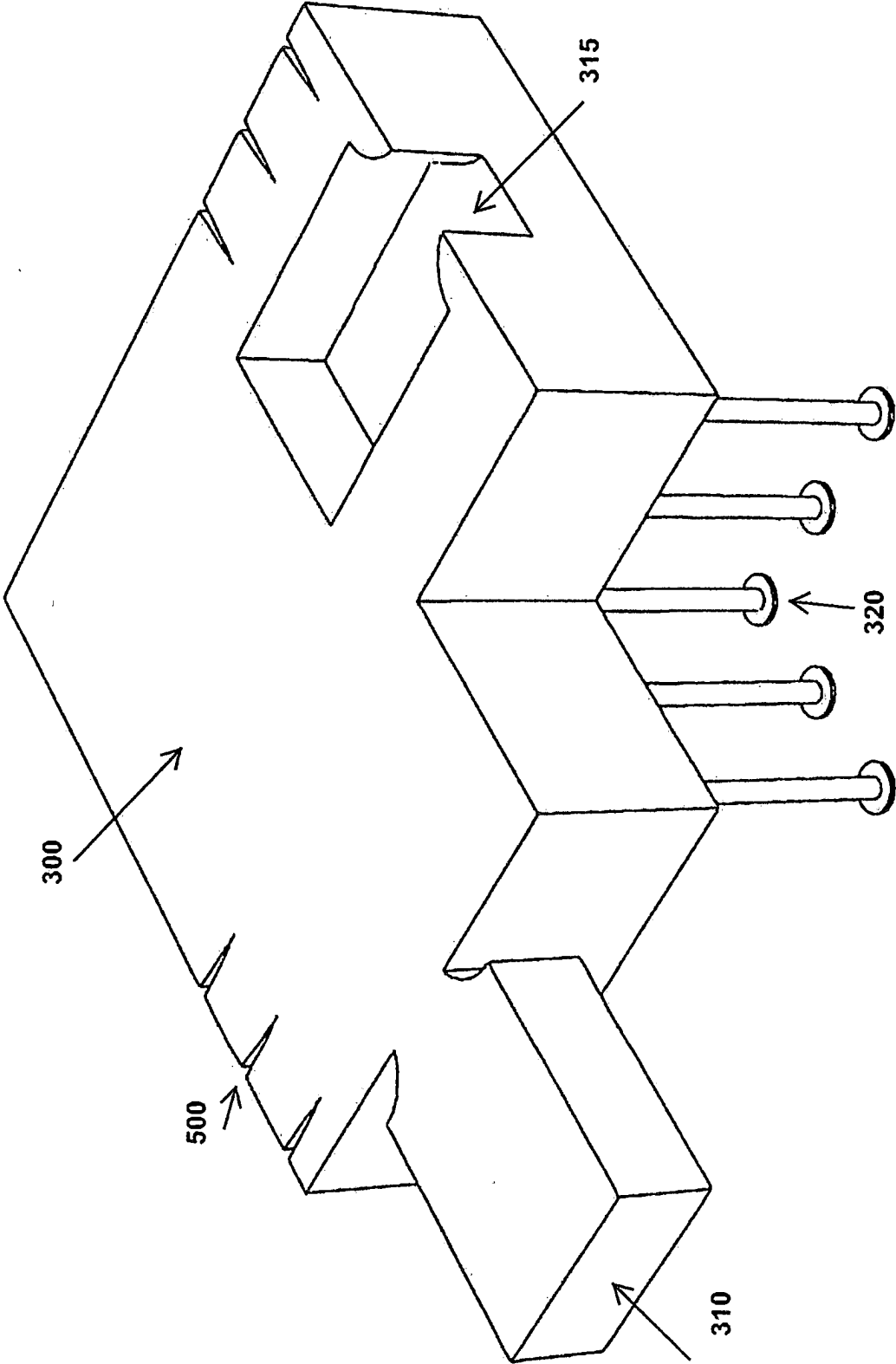


FIG. 1A

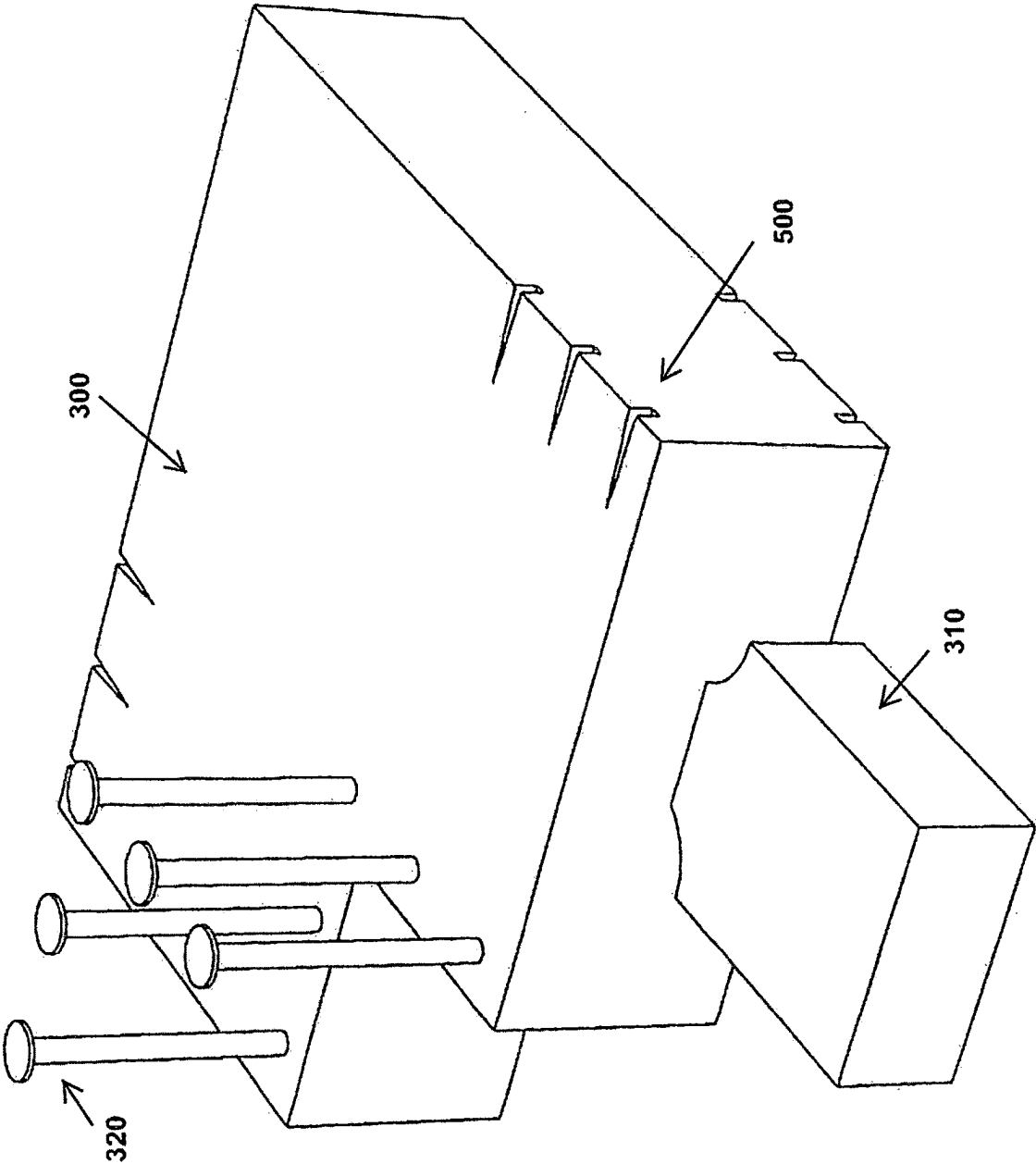


FIG.1B

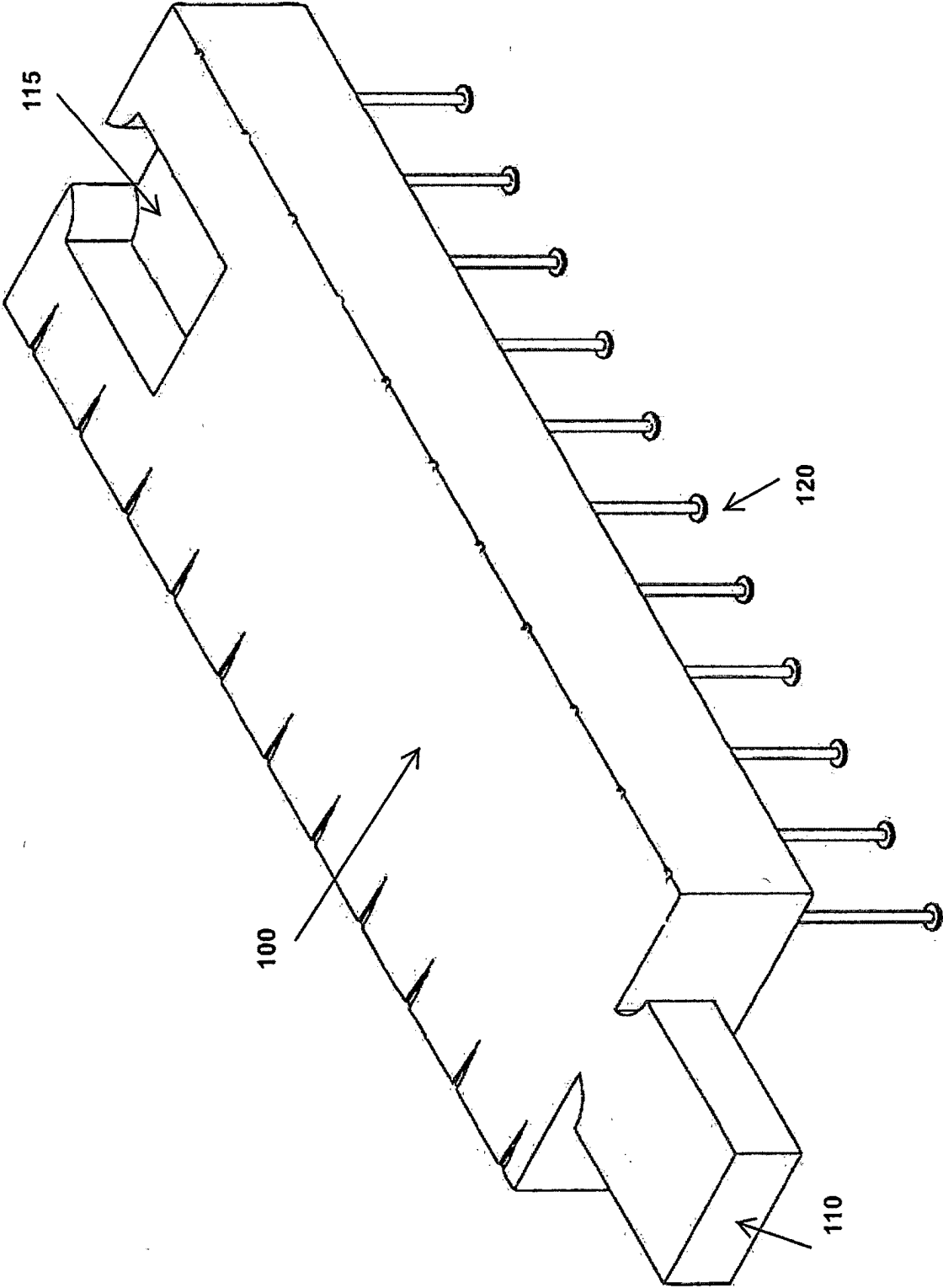
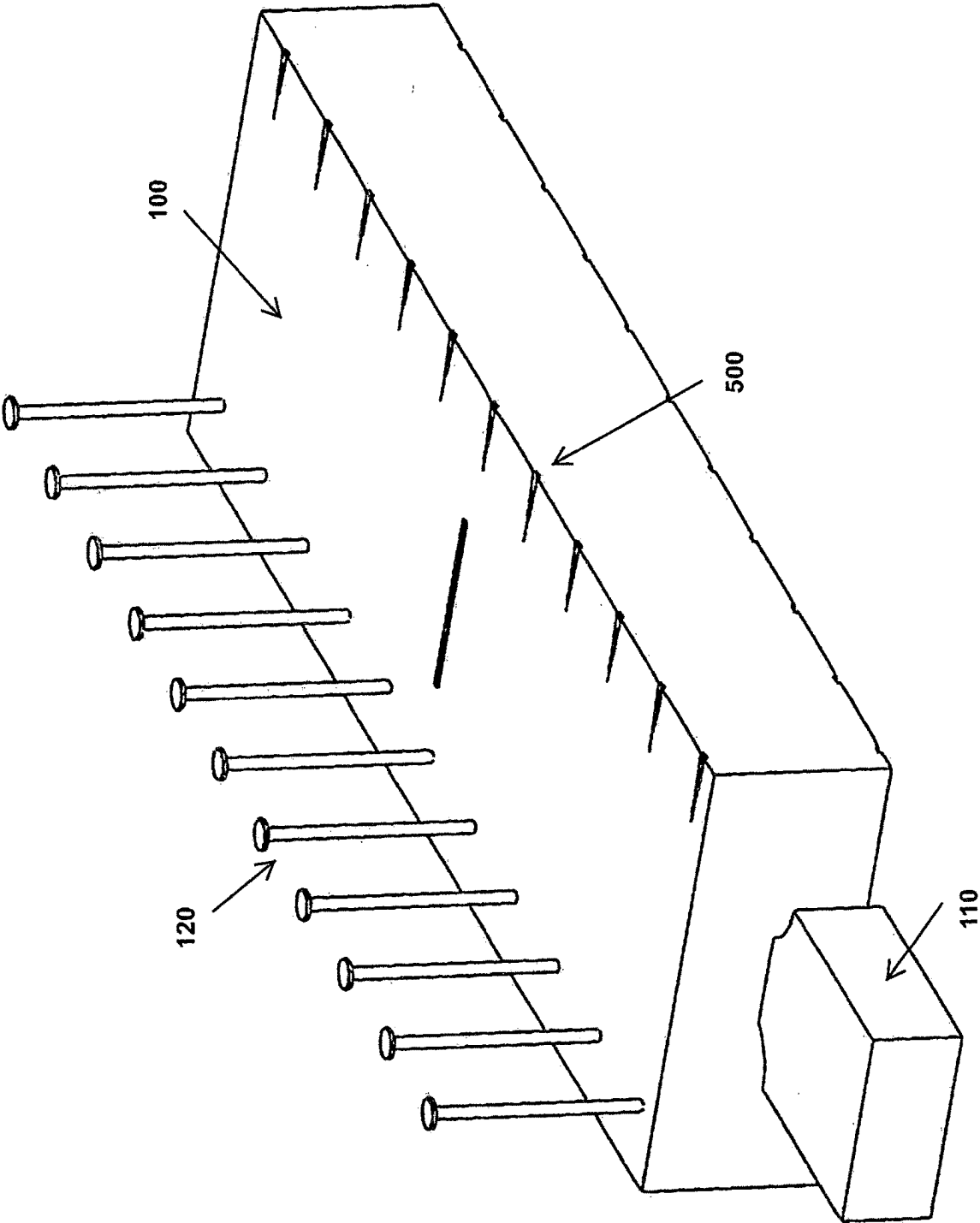


FIG. 2A

FIG. 2B



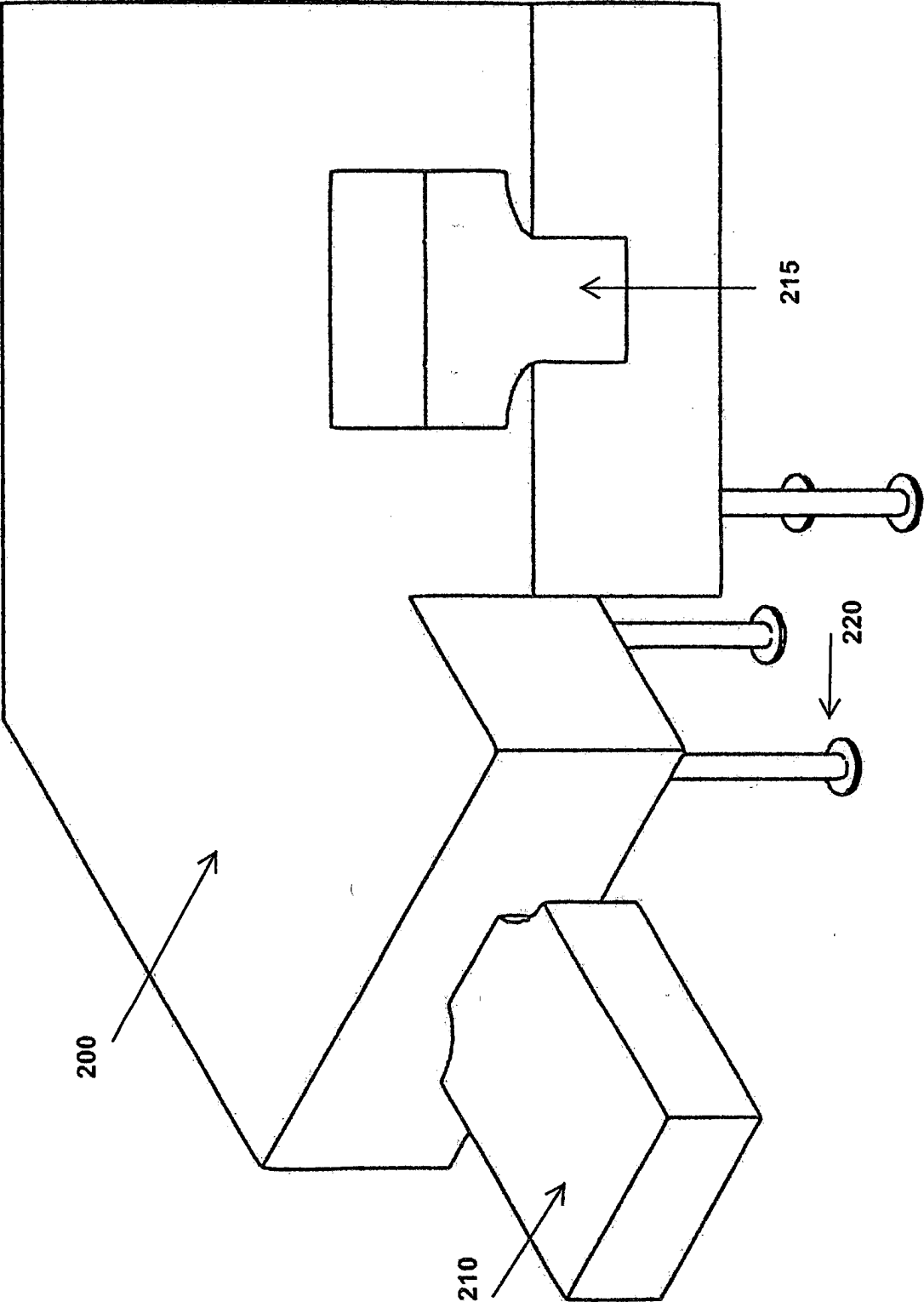


FIG. 3A

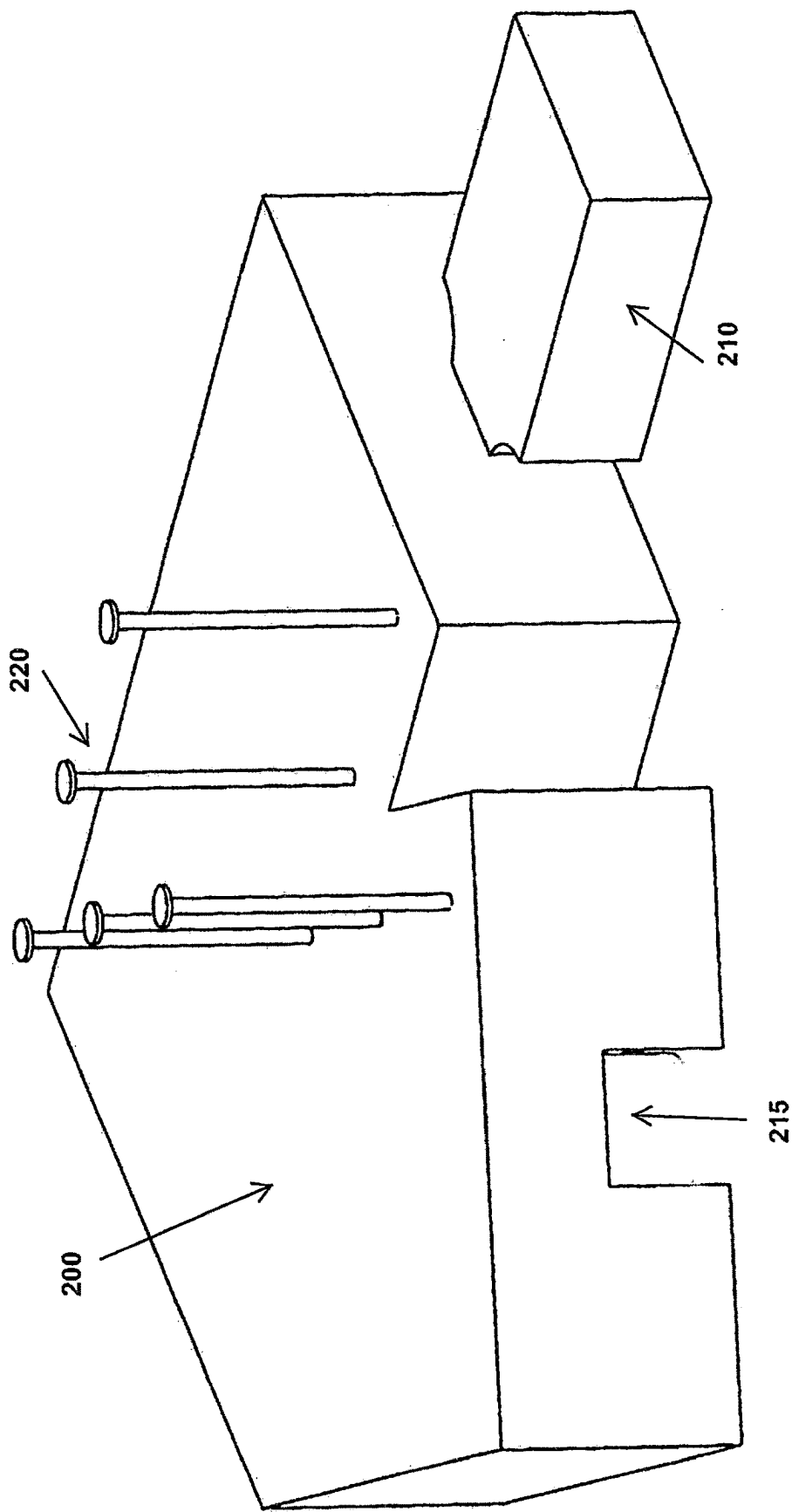
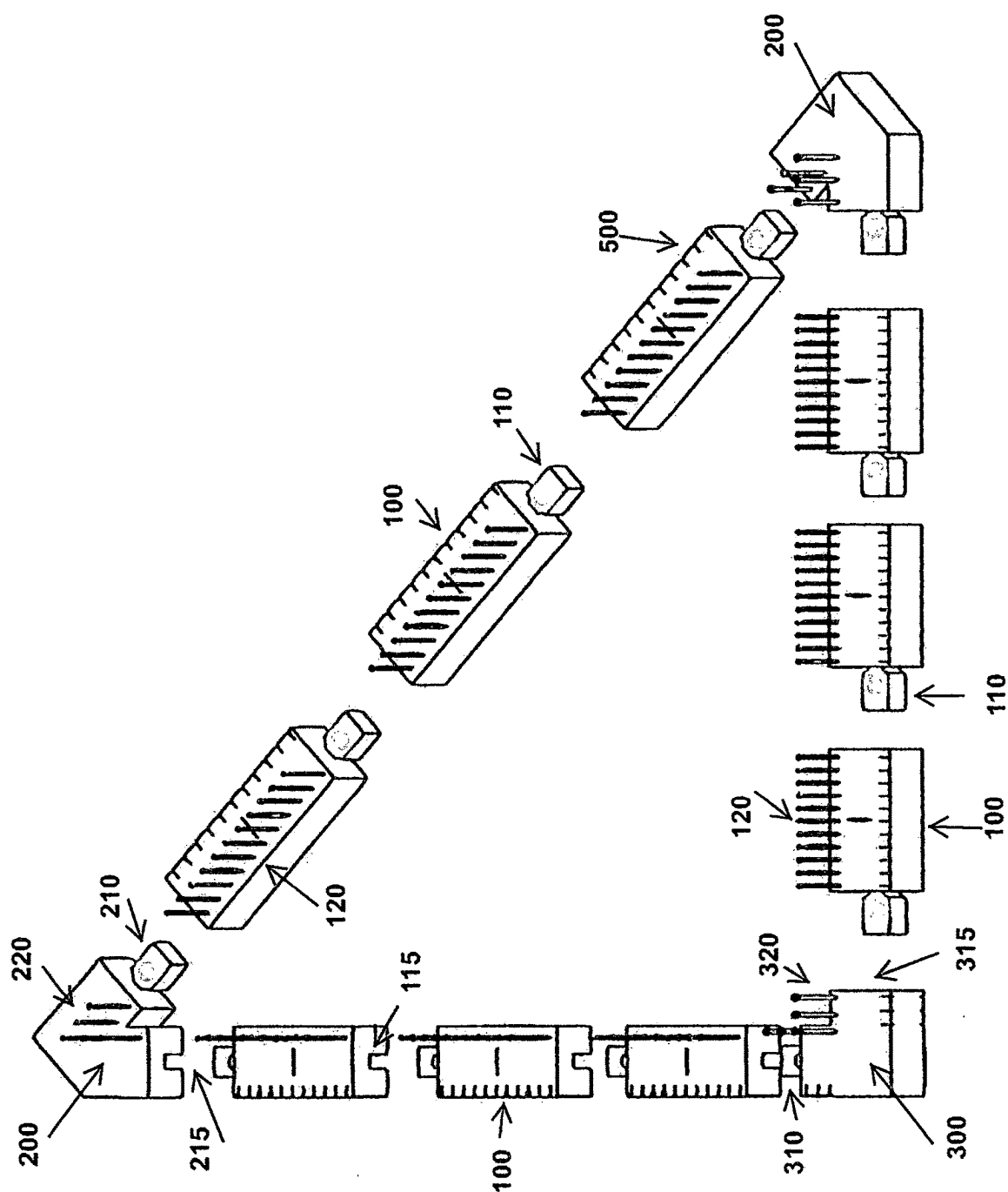


FIG. 3B

FIG-4A



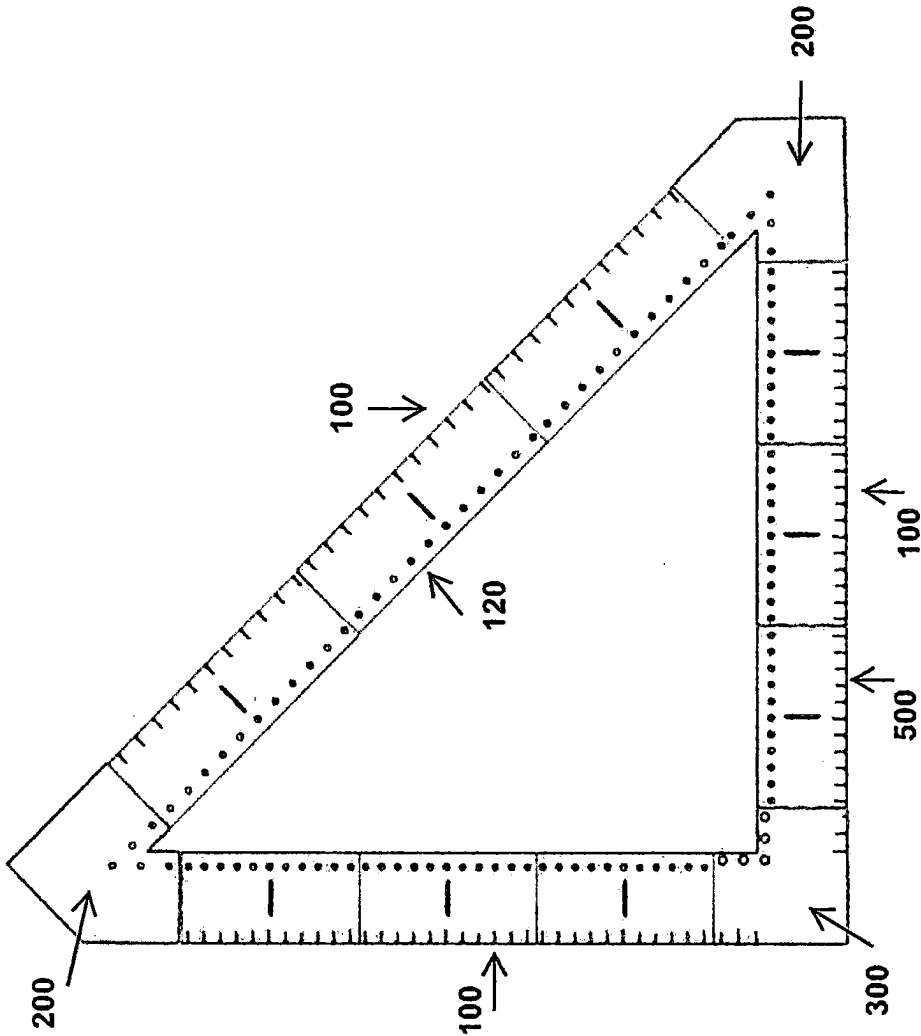


FIG. 4B

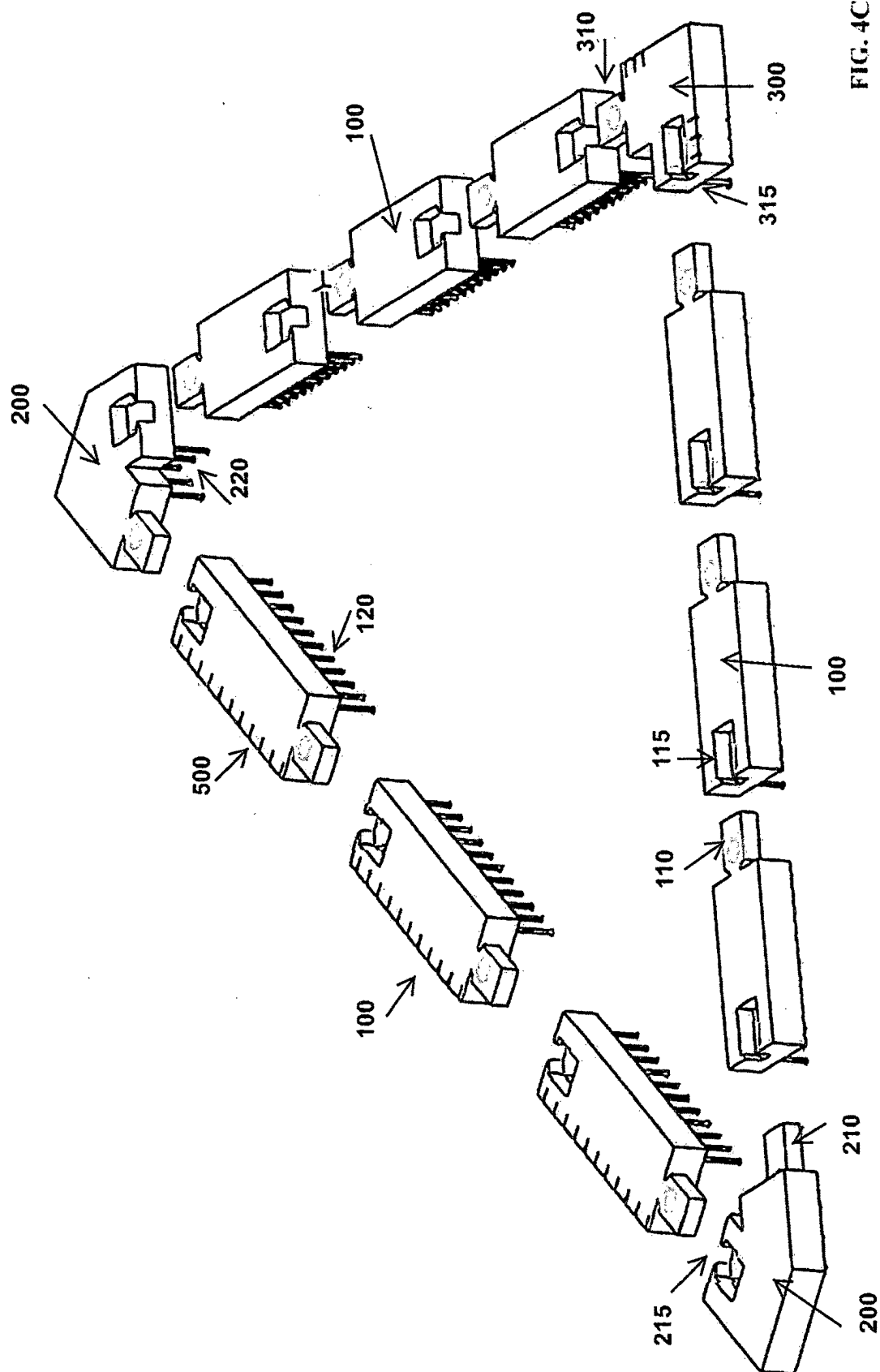


FIG. 4C

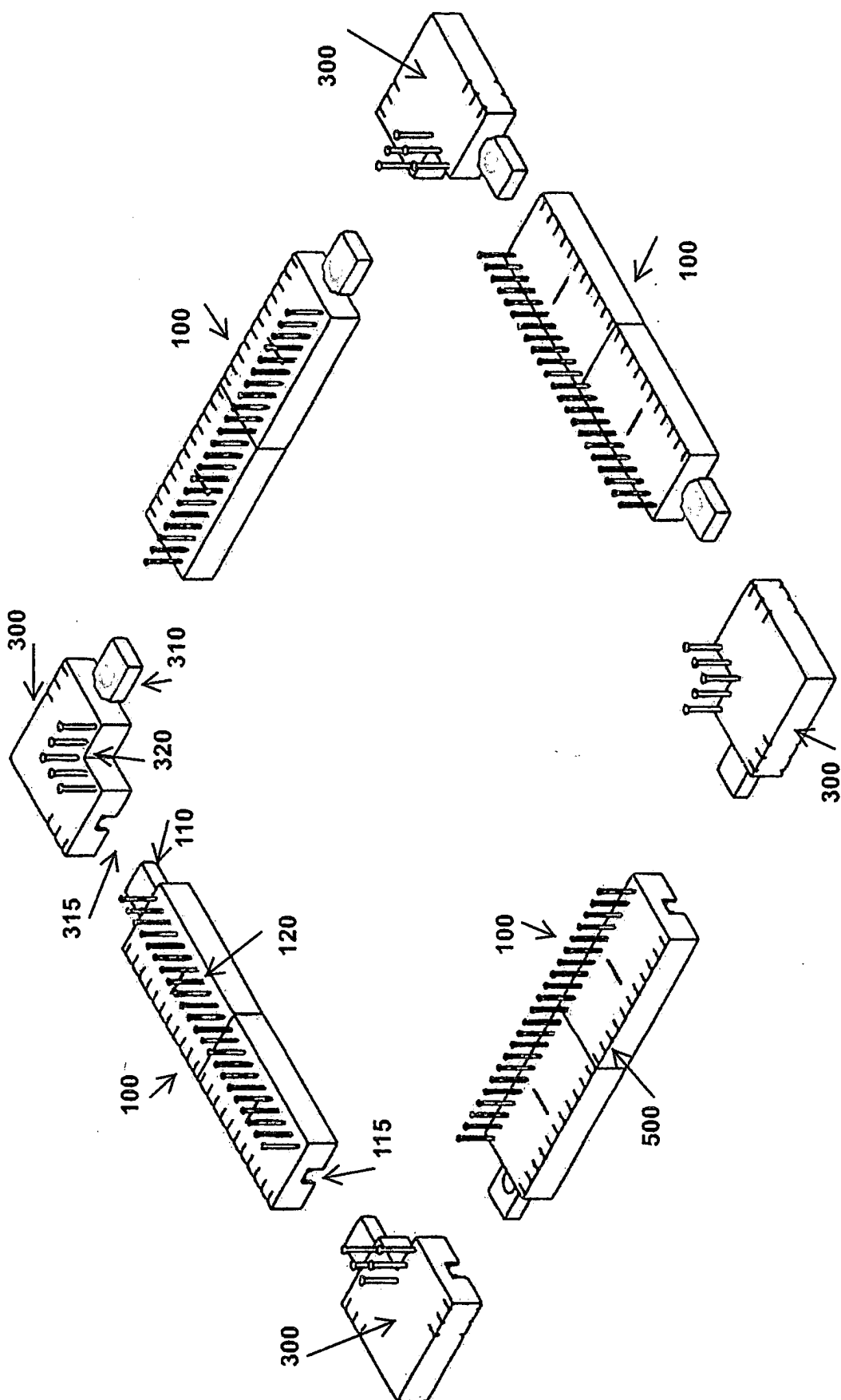


FIG. 5

MODULAR LOOM FOR TRIANGULAR OR RECTANGULAR WEAVING

FIELD OF THE DISCLOSURE

[0001] The present invention relates to a modular pin loom system for continuous strand weaving uses interlocking, reconfigurable molded plastic modules with integrated pins and connection pieces to create adjustable rectangular or triangular looms.

BACKGROUND

[0002] Continuous strand weaving on pin looms has been practiced for centuries across many cultures as an efficient, portable method of creating woven fabrics and tapestries. Early triangular and rectangular small looms were constructed of wood, using nails or pegs as the pin components around which threads were woven by hand in various patterns. These looms ranged in size from 12 inches to several feet on each side. While functional, they had limitations in terms of adjustability, portability and ease of use.

[0003] Wooden pin looms have traditionally been custom built to desired dimensions and weaving capacities. However, once constructed the dimensions are fixed, limiting the width of fabrics that can be produced. The wooden materials, while strong, add excessive weight and bulkiness to looms of increasing size. A full 8 foot long triangular or rectangular loom requires significant storage space and can be difficult to transport. With larger sizes, the fixed wooden joined pieces may also be more prone to warping or distorting during use, impacting weave quality.

[0004] Some attempts have been made to create adjustable pin looms, but these rely on interlocking wooden rails and posts of set lengths and pivot positions. Adjustability is incremental at best and rapid size or shape changes cannot be accommodated. None provide interchangeability between triangular and rectangular constructions. Additionally, excess material outside the woven area created by the adjusted intersections must be dealt with during the weaving process.

[0005] It is evident that while hand weaving on portable pin looms has a long cultural history, loom designs have several deficiencies. The custom built wooden looms with fixed dimensions common today limit adjustability in fabric widths and loom sizes. Large sizes prove bulky, heavy and awkward for storage and transport. Incremental solutions still lack adaptability, interchangeability and have design issues to resolve. Therefore what is needed is an integrated pin loom system with modular, reconfigurable pieces that eliminates these deficiencies while retaining the benefits of hand weaving on portable pin looms.

SUMMARY

[0006] In one aspect, the present invention describes a modular system for constructing customizable pin looms for continuous strand weaving of various shapes and sizes. The invention utilizes a set of interlocking molded plastic modules with integrated pins that can be assembled into adjustable triangular or rectangular frames.

[0007] One aspect describes the modular loom system containing generic side modules available in different lengths plus specialized 45 and 90 degree angle end pieces. The modules feature an integrated flange and hole system that enables secure, flush interconnections without gaps for

loom integrity and dimensional stability during weaving. Side modules are further injection molded or additive manufactured with customizable embedded pin spacing, density and notching features tailored for triangle hypotenuses or rectangular sides to enhance weaving efficiency and ergonomics at any constructed loom dimension up to 8 feet.

[0008] Another aspect describes interlocking the modules using the flanges and holes, looms ranging from 12 inches to 8 foot lengths that can be constructed by adding or subtracting modules suited for projects from small swatches to large tapestries. The ability to combine both rectangular and triangular shaping using the same modular components provides financial savings over fixed wooden looms with dedicated shaping. The molded precision pieces store efficiently and shipping costs reduce compared to bulky, awkward sized pre-fabricated pin looms.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1-3B are a top view of the interlocking pieces showing a straight side module (FIG. 2B), a 45-degree end module (FIG. 3B), and a 90-degree corner module (FIG. 1B), a plurality of pins is shown on the top surfaces of the modules.

[0010] FIG. 1-3A are a bottom view of the interlocking pieces showing the flange on each module to interlock with a corresponding flange on other modules. 90-degree corner (FIG. 1A), straight side module (FIG. 2A), and a 45-degree end module (FIG. 3A).

[0011] FIG. 4A is a top view layout of a pre-assembled triangular loom.

[0012] FIG. 4C is a bottom view layout of a pre-assembled rectangular loom.

[0013] FIG. 4B is a top view layout of an assembled triangular loom without pins installed.

[0014] FIG. 5 is a top view layout of a pre-assembled rectangular loom.

DETAILED DESCRIPTION

[0015] The exemplary embodiment according to FIGS. 1-3B illustrates a top view of three module types of the modular pin loom system interlocked together. These include a side module (100), a 45-degree end module (200), and a 90-degree corner module (300).

[0016] The side module (100) is a molded piece with an elongated rectangular shape. Pins (120) for wrapping threads are evenly spaced along the top flat surface. The pin spacing may correspond to a desired density of woven fabric to be produced. At one end of the side module (100) is a narrow, protruding flange (110).

[0017] The 45-degree end module (200) also contains an array of evenly spaced winding pins (220) spaced in accordance with the side modules. At one edge of the triangle shaped piece is an integrated flange (210). The opposite edge features a receiving hole (215) to accept the flange from another interlocking piece.

[0018] The 90-degree corner module (300) forms perpendicular sides with a flange (310) on one end. On the opposite end features a hole (315) to fit with the flanges of other modules. Winding pins (320) continue the established spacing patterns.

[0019] The provision of interlocked flanges and corresponding holes connect the modules firmly together while allowing reconfiguration as needed during the weaving process or for storage.

[0020] On the other hand, FIGS. 1A, 2A & 3A shows a bottom view of the three module types revealing the interlocking flange and hole mechanisms. The side module (100) has a protruding flange (110) centered along one short edge on the bottom surface. On the opposite short edge of the side module (100) is a narrow hole (115) to receive the flange (110) from an adjacent module to create a firm interlocked connection.

[0021] The two triangular 45-degree end modules (200) each have an integrated flange (210) running the width of the piece along the one inner edge. This mates with the receiving hole on a side module (215). The module also features a rectangular receiving hole (215) and pins (220).

[0022] Also on the figure, the 90-degree corner module (300) is designed with a flange (310) protruding from one end. Matching on the opposing end is a hole (315) to fit with the flanges of other modules, positioned to mesh securely with such flanges. Pins for weaving are on the side (320).

[0023] The integrated flanges and corresponding holes on the bottom side of the three module types (100, 200 and 300) allow them to be firmly connected together in various configurations to form the woven winding surface. The tight interlock prevents movement or distortion during the weaving process while still enabling disassembly when required.

[0024] FIGS. 4A, B & C and FIG. 5 illustrate examples of triangular and rectangular loom configurations that can be assembled from the interconnectable pin loom modules.

[0025] FIG. 4A depicts a partially constructed triangular loom layout prior to interlocking the pieces. Three elongated side modules (100) with evenly spaced winding pins (120) form the lower and left edges. The edges have a length corresponding to a desired size of a finished woven triangular panel. One 45-degree end module (200) sits adjacent to the right ends of the bottom side modules (100). The triangular shape is completed with a perpendicular side module (100) abutting the top of the 45-degree module. Finally, a 90-degree corner module (300) caps the apex opposite the triangle's "hypotenuse" edge.

[0026] Similarly, FIG. 5 shows an unfinished rectangular frame made of pin loom modules. Two longer equal-length side modules (100) lay parallel, with shorter perpendicular modules (100) not yet interlocked across the top and bottom. Four 90-degree corner modules (300) with pins (320) wait next to the open ends to complete the corners and form the final production loom once properly assembled and connected.

[0027] Once the desired loom shape and dimensions are laid out, the user simply pushes together the flanges and receiving holes on abutting modules to create a sturdy winding frame for woven fabric creation.

[0028] FIG. 4B shows the modular pin loom pieces fully interlocked into a complete triangular loom ready for fabric weaving.

[0029] The triangular loom in FIG. 4A has a plurality of modules (100) firmly connected together edge-to-edge on its three side via the protruding flanges into the mating holes. The flange (210) on the inner edge of the 45-degree end module (200) integrates seamlessly with the receiving hole (115) on the side module at the hypotenuse end. This forms a smooth, non-warping triangular frame with winding pins

(120) spaced evenly the entire perimeter for consistent threads during the weaving process. The 90-degree corner module (300) completes the apex with its two perpendicular flanges locking into place.

[0030] Similarly, FIG. 5 displays a rectangular loom with side modules (100) interlocked along the top and bottom edges by a plurality of modules connected end-to-end. The protruding flanges integrate securely into the holes on each piece to prevent separation during weaving. 90-degree corner modules (300) complete the four integrated corners of the production loom. The result is a sturdy rectangle with evenly spaced pins along all sides to facilitate efficient threading and pattern creation on the fabrics being produced.

[0031] The complete triangular and rectangular looms demonstrate how the modular system once fully assembled forms rugged, appropriately sized frames for continuous strand weaving. Further, disassembly after finishing woven items is simple and rapid for compact storage.

[0032] One aspect of the disclosure describes a series of small guides (500) to aid in keeping weaving threads organized and tangle-free, as shown in FIGS. 2A and 2B. The guides (500) are created along the outer long edges of the side modules (100), the 45-degree end module (200), and the 90-degree corner modules (300), alternating with the knobbed winding pins (120) that project from the top flat surface of each module. The open gap in the guide allows a weft thread to pass through freely to the pins across to the adjacent module edge. The tunnel forming the guide, however, captures any excess tails of material to prevent them from drooping and interfering with the woven fabric creation process. The thread instead remains up and out of the way within the guide loop pathway.

[0033] One aspect of the disclosure describes side modules (100) that make up the edges of the triangular or rectangular loom being manufactured in distinct colors—for example, blue and yellow. Such alternating of colors on the opposite side modules creates a pattern where opposite and corresponding windings pins (120) across the open loom area are easily identifiable by their matching colors. For instance, pins across from each other on the blue and yellow side modules would both be blue, then a pair of yellow pins would be opposite each other on the loom and so forth down the row. This color coding helps the weaver track the corresponding pin for each loop of thread. As the warp threads are wrapped continuously down and across the loom in sequence to create the pin woven fabric, the bold colors aid efficiency by making it easy to spot the precise opposite pin without losing one's place.

[0034] One aspect describes a single kit to construct looms ranging from 12 inches up to 8 feet along each edge simply by adding or removing modules to achieve the dimensions desired. This provides flexibility to size the loom accurately for different projects instead of being limited by fixed size wooden looms requiring separate costly purchases for incremental size changes.

[0035] In one aspect, incorporating both 45-degree angle modules (200) and 90-degree corner angle modules (300) in the modular components, users can construct both triangular and rectangular looms utilizing the same standardized side modules (100).

[0036] In one aspect, individual modules are designed not to exceed a few feet in maximum length or width. This compact size allows them to be packaged efficiently compared bulkier prefabricated 6-8 foot singular loom pieces for

shipping. Users simply interconnect matching modules upon delivery to assemble the custom sized loom for immediate use rather than housing long loom spans for storage and transport.

[0037] One aspect describes an injection molding fabrication process enables engineering each unique side module **(100)** to have customized embedded pins spacing to achieve appropriate density along both the hypotenuse and side edges in triangular loom configurations.

[0038] One aspect describes modules that may be molded with different gauge pins and hole densities allowing users to construct both fine gauge looms for delicate materials weaving and chunky gauge looms using the same modular components.

[0039] One aspect describes optional notches or grooves that may be incorporated adjacent to the embedded pin heads which enable securing the ends of the woven threads without repetitive tying and untying of knots around the pins while the fabric progresses.

[0040] One aspect describes an interlocking design where modules align precisely flush without excess overhanging wooden edges common in length adjustable wooden looms.

[0041] One aspect describes side modules **(100)** provided in varied lengths such as 6 inches and 12 inches allows more assembly combinations utilizing fewer longer pieces supplemented with shorter ones for cost efficiency.

[0042] While some aspects of the disclosure describe injection molding, the modular pin loom system components may be manufactured using other fabrication methods such as additive manufacturing (3D printing), CNC machining, or casting. Additive manufacturing techniques like fused deposition modeling (FDM) or stereolithography (SLA) can build up the modules layer-by-layer from thermoplastic filaments or UV-curable resins respectively.

[0043] In other aspects, to enhance stability and prevent distortion or biasing of the assembled loom frame, especially in larger rectangular configurations, the modular system can incorporate optional stabilizer bars. The number and length of stabilizer bars can be selected based on the overall size of the target woven item.

What is claimed is:

1. A modular pin loom system for weaving comprising: a plurality of injection molded rigid plastic modules each having embedded pins protruding from a flat surface of the module; wherein the plurality of modules include different modules including:
 - side modules in a range of lengths;
 - angle end modules; and
 - corner modules; wherein each module further comprises an integrated interlocking mechanism having either flanges or receiving holes for securely connecting to adjacent modules without gaps; wherein side modules feature customizable spacing of embedded pins to match desired loom dimensions when multiple modules are interconnected; whereby various combinations of side, angle, and corner modules can be interconnected edge-to-edge to construct triangular and rectangular shaped pin looms in sizes ranging from 12 inches to 8 feet per side.
2. The modular pin loom system according to claim 1, wherein the side modules are produced in at least two different colors to aid in weaver visualization of corresponding edge pins across assembled loom.
3. The modular pin loom system according to claim 1, wherein the rigid injection molded plastic provides non warping performance during weaving and stable precise positioning of embedded pins in matched modules.
4. The modular pin loom system according to claim 1, wherein the side modules feature optional notches adjacent to embedded pins on the flat surface to capture and secure weaving thread ends without knotting around pins.
5. The modular pin loom system according to claim 1, wherein the customizable spacing of pins provides fine to chunky gauges within the same loom construction kit to support weaving diverse materials.
6. The modular pin loom system according to claim 1, wherein the flanges and receiving holes enable flush aligned interconnections between all modules, eliminating uneven edges or overhang.
7. The modular pin loom system according to claim 1, wherein the modular components with embedded pins are manufactured utilizing precision automated processes for reliability and reproducibility at high volumes.

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